

CSE 370– Database Systems

QUIZ 3 (Set A)

Spring' 25

Marks

Full Name (in Block Letter): _____

ID: _____ **Section:** ____ **Signature:** _____

Date: _____ **Total Marks: 10** **Time: 10 minutes**

Consider the following relation:

Music_Stream (platformID, trackID, playID, sessionID, platformName, subscriptionType, launchYear, headquarters, subscriptionFee, genre, duration, trackTitle, artistID, artistName, artistEmail, listenerRating, playCount)

Primary Key: (platformID, trackID, playID, sessionID)

Functional Dependencies:

- **FD1:** artistID → artistName, artistEmail
- **FD2:** subscriptionType, launchYear → subscriptionFee
- **FD3:** platformID, trackID, playID, sessionID → platformName, subscriptionType, launchYear, headquarters, subscriptionFee, genre, duration, trackTitle, artistID, artistName, artistEmail, listenerRating, playCount
- **FD4:** trackID → genre, duration, trackTitle
- **FD5:** platformID → platformName, subscriptionType, launchYear, headquarters, subscriptionFee
- **FD6:** trackID, playID, sessionID → artistID, artistName, artistEmail

(a) Explain if the above relation is in the **first normal form (1NF)** or not. If not, apply **1NF normalization**.

Answer: There are no multivalued attributes, composite attributes or nested relations in the above table. So it is in 1NF.

(b) Explain if the relation(s) from part (a) is/are in the **second normal form (2NF)** or not. If not, apply **2NF normalization**.

Answer: In FD4, FD5 and FD6 **non-prime attributes are partially** dependent on a **subset** of this composite key, resulting in partial dependencies. Hence, relation from part a is not in 2NF.

Decomposition:

Platform(platformID, platformName, subscriptionType, launchYear, headquarters, subscriptionFee)

← from FD5

Track(trackID, genre, duration, trackTitle)

← from FD4

Artist_Map(trackID, playID, sessionID, artistID, artistName, artistEmail)

← from FD6

Music_Stream_Main(platformID, trackID, playID, sessionID, listenerRating, playCount)

(c) Explain if the relation(s) from part (b) is/are in the **third normal form (3NF)** or not. If not, apply **3NF normalization**.

Answer: Due to **transitive dependencies** (non-prime attribute depending on another non-prime attribute) in FD1 and FD2, the relations from part b are not in 3NF.

Decomposition:

Platform(platformID, platformName, subscriptionType, launchYear, headquarters)

Subscription(subscriptionType, launchYear, subscriptionFee)
← from FD2

Track(trackID, genre, duration, trackTitle)

Artist(artistID, artistName, artistEmail)
← from FD1

Artist_Map(trackID, playID, sessionID, artistID)

Music_Stream_Main(platformID, trackID, playID, sessionID, listenerRating, playCount)